

Linear-Time Efficiency and Physical Manifold Fidelity in Symplectic Selective State Space Models

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Abstract

We present the **Physically-Grounded Selective State Space (PG-S3)** model, a world-model architecture that simultaneously achieves $O(L)$ computational complexity and high fidelity to the underlying physical manifold of modeled systems. By replacing the unconstrained Zero-Order Hold (ZOH) transition used in selective state space models with a Störmer–Verlet symplectic integrator, and encoding the latent state as a Hamiltonian phase-space vector $h = [q, p]^\top$, PG-S3 preserves geometric structure throughout long autoregressive rollouts. We further augment the core with (i) a neural-physical coupling layer whose phase-space transform satisfies the symplecticity constraint $W^\top J W = J$ by construction, (ii) a branchless Gershgorin CFL stability bound on the adaptive time step Δ_t , and (iii) Scale-Adaptive INT4 Quantization (SAQ) targeting 8 GB CPU-bound hardware. A five-phase empirical campaign validates the framework end-to-end: production runs sustaining 1.2×10^6 steps per second across 524 288 trajectories achieve a total fractional Hamiltonian drift of $1.26 \times 10^{-11} E_0$ within a peak resident-set size of 4 096.5 MB. A controlled harmonic-oscillator audit yields $ECE_{\text{rms}} = 1.4886 \times 10^{-5} \approx ECE_{\text{max}} / \sqrt{2}$, the sinusoidal identity expected when the trajectory remains on a nearby shadow energy surface. Topological phase-space reconstruction via the Rosenstein algorithm yields a maximal Lyapunov exponent $\lambda_{\text{max}} = 0.3578$ ($R^2 = 0.9792$) and a predictability horizon $L_{\text{crit}} = 2.79$.

1 Introduction

World models—learned simulators that predict future states through latent “imagination”—underpin modern model-based reinforcement learning, physical simulation, and autonomous reasoning [1, 2]. Yet the dominant Transformer-based implementations incur $O(L^2)$ time and memory complexity with respect to sequence length L , making long-horizon rollouts on consumer-grade, CPU-bound hardware prohibitive.

Selective State Space Models (SSMs) such as Mamba [3] recover $O(L)$ complexity via a recurrent scan, but their standard Zero-Order Hold (ZOH) discretization is not constrained to be Hamiltonian or symplectic in canonical phase-space coordinates. The exact exponential of a

Hamiltonian linear system is symplectic, but the generic input-dependent affine transition used by selective SSMs need not respect the canonical two-form. This mismatch can introduce artificial energy drift over long rollouts, systematically degrading predictive fidelity in physics-rich environments.

Independently, Hamiltonian Neural Networks [4] and Symplectic Neural Networks [5] have demonstrated that enforcing Hamiltonian structure improves long-horizon accuracy, but have not been integrated into the efficient linear-scan paradigm or hardware optimization strategies required for CPU deployment.

Contributions. This paper closes that gap through three primary contributions:

1. **Symplectic Selective Dynamics:** We replace ZOH with a Störmer–Verlet integrator inside the SSM scan, provably preserving the symplectic 2-form of the latent Hamiltonian flow.
2. **Hardware-aware CPU deployment:** An AVX2-optimized branchless kernel with SAQ INT4 quantization executes the $O(L)$ scan in under 420 ms for $L = 16\,384$ with peak RSS well below 5 GB.
3. **End-to-end empirical validation:** Five verification phases—complexity benchmarking, energy conservation, neural-physical coupling, production-scale profiling, and topological reconstruction—confirm both theoretical properties and practical hardware constraints simultaneously.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 develops the mathematical framework. Section 4 describes the architecture and hardware implementation. Section 5 presents the empirical evaluation. Section 6 discusses limitations, and Section 7 concludes.

2 Related Work

Selective State Space Models. Mamba [3] introduces input-dependent selection into linear SSMs, achieving competitive performance with Transformers at $O(L)$

complexity via a hardware-aware parallel scan. Subsequent work (H3 [6], Hyena [7], S4 [8]) focuses on expressiveness and long-range dependencies, but none addresses physical conservation laws or CPU-bound deployment.

Physics-Informed Neural Architectures. Hamiltonian Neural Networks [4] learn energy-conserving dynamics by parameterising $\mathcal{H}(q, p)$ as a network. Symplectic Recurrent Neural Networks [5] learn Hamiltonian dynamics while unrolling with symplectic integration. Neural ODEs [9] embed continuous dynamics into the latent space but do not pair this with the efficient recurrent scan needed for $O(L)$ inference. The combination of symplectic latent dynamics, selective SSM-style scanning, 4-bit quantization, and register-level SIMD execution is not the focus of these methods.

CPU Inference Optimization. GGUF/llama.cpp [10] demonstrates practical INT4 deployment on commodity CPUs, but is not designed for physically-constrained dynamics. OpenVINO [11] provides a custom-op extension framework that we leverage for hardware-specific kernel registration. FAVOR+ [12] approximates attention with $O(ndr)$ random-feature kernels; we adapt it as the linear-attention baseline in our benchmarking study.

Lyapunov Analysis. The Rosenstein algorithm [13] estimates the maximal Lyapunov exponent from a single time series via delay embedding and nearest-neighbour divergence tracking; we employ it in Phase 05 to characterise the attractor of the learned latent dynamics.

3 Mathematical Framework

3.1 Selective State Space Models

A continuous-time selective SSM evolves as

$$\dot{h}(t) = \mathbf{A} h(t) + \mathbf{B} x(t), \quad (1)$$

$$y(t) = \mathbf{C} h(t), \quad (2)$$

where $h(t) \in \mathbb{R}^d$ is the latent state, $x(t) \in \mathbb{R}^n$ is the input, and the selective parameters $\mathbf{A}_t, \mathbf{B}_t, \mathbf{C}_t$ are functions of the current token. The standard ZOH discretization yields $\tilde{\mathbf{A}} = \exp(\Delta \mathbf{A})$, $\tilde{\mathbf{B}} = \int_0^\Delta \exp(\tau \mathbf{A}) d\tau \mathbf{B}$, which reduces to $\mathbf{A}^{-1}(\exp(\Delta \mathbf{A}) - I)\mathbf{B}$ when $\mathbf{A}$ is nonsingular. The recurrence $h_t = \tilde{\mathbf{A}} h_{t-1} + \tilde{\mathbf{B}} x_t$ is parallelised via an associative prefix scan in $O(L)$ work and $O(\log L)$ span [14].

3.2 Symplectic Selective Dynamics

Hamiltonian Partitioning. We partition the latent state as $h = [q, p]^\top$ with $q, p \in \mathbb{R}^{d/2}$ representing gener-

alised coordinates and momenta. The latent Hamiltonian is $\mathcal{H}(q, p) = T(p) + V(q)$, where T and V are the kinetic and potential energy components learned by the neural network.

Why the standard selective transition is insufficient. ZOH treats $\mathbf{A}_t, \mathbf{B}_t$ and the forcing $\bar{\mathbf{B}}_t x_t$ as fixed over $[t, t + \Delta]$. This is numerically legitimate for generic linear systems, but it does not enforce the Hamiltonian block structure $\dot{h} = J \nabla \mathcal{H}(h)$ or the discrete condition $M_t^\top J M_t = J$ on the step Jacobian M_t . Thus a learned selective transition may expand or contract canonical phase-space volume and accumulate nonphysical Hamiltonian error unless the transition is explicitly constrained.

Störmer–Verlet Replacement. We replace ZOH with a second-order, time-reversible, symplectic integrator. Given $\nabla_q V$ and $\nabla_p T$ from the network, one step is:

$$p_{t+1/2} = p_t - \frac{\Delta}{2} \nabla_q V(q_t), \quad (3)$$

$$q_{t+1} = q_t + \Delta \nabla_p T(p_{t+1/2}), \quad (4)$$

$$p_{t+1} = p_{t+1/2} - \frac{\Delta}{2} \nabla_q V(q_{t+1}). \quad (5)$$

Equations (3)–(5) define the *Symplectic Selective Update* that replaces the ZOH kernel.

Proposition 1 (Symplecticity of the PG-S3 core). *For a separable latent Hamiltonian $\mathcal{H}(q, p) = T(p) + V(q)$, the map $(q_t, p_t) \mapsto (q_{t+1}, p_{t+1})$ defined by Eqs. (3)–(5) is symplectic: its Jacobian M satisfies $M^\top J M = J$, where $J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$.*

Proof. The first and third updates are exact flows of the potential Hamiltonian $V(q)$ for a half step, and the middle update is the exact flow of the kinetic Hamiltonian $T(p)$ for a full step. Exact Hamiltonian flows are symplectic, and a composition of symplectic maps is symplectic. $\square$

Consequently, Liouville’s theorem holds for the conservative core. The numerical trajectory exactly conserves a modified, or shadow, Hamiltonian

$$\tilde{\mathcal{H}} = \mathcal{H} + O(\Delta^2), \quad (6)$$

over exponentially long intervals under the usual smoothness and boundedness assumptions for backward error analysis [15].

3.3 Gershgorin CFL Stability Condition

For a latent transition matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, the Gershgorin Circle Theorem bounds its spectral radius as

$$\rho(\mathbf{A}) \leq \max_i \sum_j |a_{ij}| =: \rho_G. \quad (7)$$

Stability of the Störmer–Verlet step requires $\omega\Delta_t \leq 2$ for a harmonic mode of angular frequency ω . Using ρ_G as a conservative upper bound on the largest resolved frequency yields the *Latent CFL Condition*:

$$\Delta_t = \min\left(\Delta_{\max}, \frac{2\eta}{\max(\rho_G, \epsilon)}\right), \quad (8)$$

$$\Delta_{\max} = 0.01, \quad 0 < \eta \leq 1.$$

where ϵ prevents division by zero and η is an optional safety factor. The factor of 2 follows from the linear stability analysis of the Störmer–Verlet map applied to a harmonic oscillator [15]. This bound is computed branchlessly via AVX2 horizontal reduction on each row of $\mathbf{A}$, contracting Δ_t during high-gradient perturbations without pipeline stalls.

3.4 Lyapunov Predictability Horizon

Chaotic physical systems exhibit exponential divergence of nearby trajectories governed by the maximal Lyapunov exponent $\lambda_{\max}$:

$$\|\delta h(t)\| \approx e^{\lambda_{\max} t} \|\delta h(0)\|. \quad (9)$$

We define the *Predictability Horizon* as the sequence length at which phase error $\|\delta h\|$ exceeds a threshold ϵ :

$$L_{\text{crit}} = \frac{1}{\lambda_{\max}} \ln\left(\frac{\epsilon}{\|\delta h(0)\|}\right). \quad (10)$$

Because the Störmer–Verlet map is symplectic for the conservative core, it follows the shadow Hamiltonian $\tilde{\mathcal{H}}$ without artificial phase-volume expansion. This does not change the true Lyapunov exponent of the target system; rather, it reduces numerical contributions to the measured divergence and makes L_{crit} a property of the modeled dynamics rather than an artifact of the integrator.

3.5 Rate-Distortion Bound for INT4 Quantization

Quantising projection weights $\mathbf{B}, \mathbf{C}$ to 4-bit integers introduces distortion $D = \mathbb{E}[\|\mathbf{W}_{\text{FP32}} - \mathbf{W}_{\text{INT4}}\|^2]$. For an ideal symmetric uniform quantizer on $[-\alpha, \alpha]$ with step size $\Delta_q = 2\alpha/(2^b - 1)$, $b = 4$, and approximately uniform rounding noise, the noise floor is $\sigma_q^2 = \Delta_q^2/12 = \alpha^2/675$.

For a Kolmogorov energy spectrum $E(k) \propto k^{-5/3}$, the tail energy beyond wavenumber $k_{\max}$ is $\int_{k_{\max}}^{\infty} E(k) dk = \frac{3}{2} C k_{\max}^{-2/3}$. The *Scale-Adaptive Quantization (SAQ)* condition $\sigma_q^2 < \int_{k_{\max}}^{\infty} E(k) dk$ requires that the quantisation scale α is adapted locally to the tile dynamic range, ensuring INT4 precision does not catastrophically filter the Kolmogorov cascade. Concretely, SAQ sets $\alpha(t) = \max_i |w_i(t)|$, recomputed per tile to track the local dynamic range of $\mathbf{B}$ and $\mathbf{C}$.

4 Methodology

4.1 Architecture Overview

An input token sequence $x_{1:L}$ is projected to $[q_0, p_0]^\top$ by the symplectic encoder (Section 4.3). The Symplectic Selective Update (Eqs. (3)–(5)) propagates the latent state while the Gershgorin CFL bound (Eq. (8)) adapts Δ_t on the fly. The output projection $\mathbf{C}$ maps $h_t \rightarrow y_t$ using SAQ INT4 weights. The full scan is implemented as a Work-Efficient Blleloch Scan [14] with hybrid tile sizes $T = 64$ on x86-AVX2 and $T = 16$ on ARM-NEON. The SAQ scale condition is enforced per tile using each tile’s local $\alpha(t)$, so the dynamic-range guarantee is applied uniformly across all weight tiles.

4.2 SIMD Kernel Specification

The innermost integration loop operates on Structure-of-Arrays (SoA) vectors $q, p \in \mathbb{R}^{d/2}$ stored in 32-byte-aligned YMM registers. The AVX2 kernel implements branchless clamping and min/max saturation:

- **VPADDD**: 32-bit accumulation.
- **VPMINSD/VPMAXSD**: saturated arithmetic.
- **VPAND/VPXOR**: logical branchless masks.
- **VFMADD231PS**: fused multiply-add for gradient updates in Eqs. (3)–(5).

All conditional logic is replaced by mask computation, e.g.:

$$\text{val} = (\text{mask} \& \text{lim}) | (\sim \text{mask} \& \text{val}).$$

SAQ INT4 is packed as 2 elements per byte with scale $\alpha(t)$ stored per 64-element tile. Dequantisation unpacks nibbles to signed bytes, then uses **VPMOVSXBD** followed by **VCVTDQ2PS**.

4.3 Neural-Physical Coupling

After an initial dimensional lift into an even-dimensional phase space, the coupling layer maps latent coordinates through a sequence of three elementary symplectic shears $W = S_3 S_2 S_1$, where each $S_i \in SL(2, \mathbb{R})$ is a shear matrix. The resulting Jacobian M satisfies $\det(M) = 1$, verified analytically:

$$\begin{aligned} q_1 &= q_0 + \alpha p_0, & p_1 &= p_0 + \beta q_1, \\ q_2 &= q_1 + \gamma p_1, & p_2 &= p_1. \end{aligned}$$

Expanding the determinant: $(1 + \gamma\beta)(1 + \alpha\beta) - \beta(\alpha + \gamma + \gamma\alpha\beta) = 1$. For one degree of freedom, $SL(2, \mathbb{R}) = Sp(2, \mathbb{R})$; hence $W^\top J W = J$. For the full d -dimensional latent space with $q, p \in \mathbb{R}^{d/2}$, symplecticity is preserved

Algorithm 1: PG-S3 Symplectic Selective Scan**Input:** sequence $x_{1:L}$; selectors for $\mathbf{A}_t, \mathbf{B}_t, \mathbf{C}_t$.**Output:** sequence $y_{1:L}$.

1. Initialize $(q_0, p_0) \leftarrow \text{SymplecticEncoder}(x_1)$.
2. For $t = 1, \dots, L$:
 - (a) Select $(\mathbf{A}_t, \mathbf{B}_t, \mathbf{C}_t)$ from x_t .
 - (b) Compute $\rho_G \leftarrow \max_i \sum_j |a_{ij}|$ and $\Delta_t \leftarrow \min(\Delta_{\max}, 2\eta / \max(\rho_G, \epsilon))$.
 - (c) Set $\hat{f}_t^p \leftarrow \text{proj}_p(\mathbf{B}_t x_t)$.
 - (d) Apply the forced Verlet half-kick: $p_{t+1/2} \leftarrow p_t - \frac{\Delta_t}{2} \nabla_q V(q_t) + \frac{\Delta_t}{2} \hat{f}_t^p$.
 - (e) Apply the drift: $q_{t+1} \leftarrow q_t + \Delta_t \nabla_p T(p_{t+1/2})$.
 - (f) Apply the final half-kick: $p_{t+1} \leftarrow p_{t+1/2} - \frac{\Delta_t}{2} \nabla_q V(q_{t+1}) + \frac{\Delta_t}{2} \hat{f}_t^p$.
 - (g) If damping is enabled, apply $p_{t+1} \leftarrow \exp(-\gamma \Delta_t) \odot p_{t+1}$.
 - (h) Emit $h_t = [q_{t+1}, p_{t+1}]^\top$ and $y_t \leftarrow \mathbf{C}_t h_t$.

by constructing W as a block-diagonal product of $d/2$ independent 2D shear pairs, each acting on a (q_i, p_i) coordinate pair. The global map satisfies $W^\top J_{d \times d} W = J_{d \times d}$ by the direct-sum structure of block-diagonal symplectic matrices [15].

Dissipative forcing (modelling non-conservative environments) is handled as an optional operator-split Rayleigh damping step rather than as part of the conservative symplectic core:

$$F_i = -\gamma_i p_i, \quad \gamma_i \geq 0, \quad (11)$$

which ensures the dissipation bound $\langle F, p \rangle = -\sum_i \gamma_i p_i^2 \leq 0$. Equivalently, the vectorized update $p_i \leftarrow \exp(-\gamma_i \Delta_t) p_i$ is conformally symplectic and monotonically removes kinetic energy when damping is enabled.

Zero-copy data ingestion is implemented via `mmap` with `MAP_SHARED`, streaming trajectories from a 1 GB pinned buffer without heap allocation during the scan.

5 Experiments

We conduct a five-phase empirical evaluation on a single CPU-only machine (Intel Core i5, 8 GB RAM, macOS Ventura, AVX2 support). All C++ kernels are compiled with `g++ -mavx2 -mfma -O3` and environment-pinned via `OV_CPU_DISPATCH_ARCH=AVX2`.

5.1 Phase 01: Complexity Analysis and Benchmarking

We benchmark the PG-S3 scan across sequence lengths $n \in \{128, 512, 2048, 8192, 16384\}$ using 20 warm-up

passes and 50 repetitions per configuration. Table 1 reports median latency and peak resident-set size (RSS) for the symplectic SSM scan. FAVOR+ Performer [12] is retained as a linear-attention reference in the harness, but PG-S3 replaces attention entirely with the recurrent symplectic scan.

Table 1: Complexity benchmarks (CPU, Intel i5, single thread). RSS measured via `psutil`.

Seq. len. n	Latency (ms)	RSS (MB)
128	6.71	26.72
512	17.40	27.83
2048	57.31	34.41
8192	215.02	60.31
16384	419.03	99.71

A log-log regression on latency versus n yields $R^2 = 0.996$ with slope 0.861 over the full measured range, and slope 0.920 when the shortest length is excluded. This is consistent with the theoretical $O(n)$ work bound, with fixed overheads and cache effects visible at small sequence lengths. Peak RSS is well within the 5 GB hardware budget at all lengths. A non-linear jump at $n = 2048$ in earlier profiling was traced to kernel dispatch overhead and does not affect the asymptotic bound.

Quantization memory footprint. For $L = 10^6$, $d = 512$, SAQ INT4: $10^6 \times 512 \times 0.5 \text{ B} = 256 \text{ MB}$ for base tensors plus $< 2 \text{ GB}$ for intermediate buffers, confirmed well below the 5 GB ceiling.

5.2 Phase 02: Symplectic Energy Conservation

We implement the Störmer–Verlet kernel (Eqs. (3)–(5)) as an OpenVINO custom operator with AVX2 intrinsics, operating on SoA $[q, p]^\top$ vectors with $m = k = 1$ (natural units). The Gershgorin CFL condition (Eq. (8)) is enforced via branchless row-sum reduction, yielding $\Delta_t = \min(0.01, 2/\omega) = 0.01$.

The Energy-Conserving Error is defined as:

$$\text{ECE}(t) = \left| \frac{\mathcal{H}(h_t) - \mathcal{H}(h_0)}{\mathcal{H}(h_0)} \right|. \quad (12)$$

Over 10^3 integration steps from initial conditions $q_0 = 1$, $p_0 = 0$ ($\mathcal{H}_0 = 0.5$):

Two properties are notable. First, $\text{ECE}_{\max} = 2.5000 \times 10^{-5}$ matches the analytic Störmer–Verlet harmonic-oscillator bound $\omega^2 \Delta t^2 / 4 = 2.5 \times 10^{-5}$ to four significant figures, confirming that the AVX2 kernel implements the Störmer–Verlet map correctly. Second, $\text{ECE}_{\text{rms}} = 1.4886 \times 10^{-5} \approx \text{ECE}_{\max} / \sqrt{2}$, the identity for a purely

Table 2: Phase 02 genuine symplectic validation ($\omega = 1$, $\Delta t = 0.01$, 1000 steps).

Metric	Value
$ECE_{\max}$	2.5000×10^{-5}
ECE_{rms}	1.4886×10^{-5}
$\mathcal{H}_{\text{final}}$	0.49999630
Bounded energy band	True

sinusoidal oscillation, confirming that the error is *oscillatory and bounded* rather than growing—the defining signature of a bounded orbit on the shadow energy surface [15]. The final Hamiltonian value $\mathcal{H}_{\text{final}} = 0.49999630$ represents a fractional drift of 3.7×10^{-6} from $\mathcal{H}_0 = 0.5$, consistent with $ECE_{\max}$.

High-gradient stress tests (Section 5.5) further confirm that ECE remains bounded under Euler integration failure modes, with the symplectic advantage growing to $268,400\times$ at 10^4 steps. This is consistent with the production energy drift of $1.26 \times 10^{-11} E_0$ observed in Phase 04, where the Gershgorin CFL condition contracts Δ_t to $\approx 5 \times 10^{-4}$, suppressing the shadow Hamiltonian offset by a further factor of ~ 400 .

5.3 Phase 03: Neural-Physical Coupling Validation

With the symplectic coupling layer integrated, we run 1000-step rollouts on 2D Navier–Stokes and rigid-body environments.

Table 3: Phase 03 coupling metrics (1000-step rollout).

Metric	Value
$ECE_{\max}$	4.82×10^{-8}
VPS (mean square curl)	1.25×10^{-9}
L_{crit} (predictability horizon)	42.1 steps
Peak RSS	4.20 GB
Streaming buffer	850 MB

The Phase 03 value $L_{\text{crit}} = 42.1$ reflects the local coupling stability of the 1000-step rollout, probing a near-integrable regime with effective $\lambda_{\max} \approx 0.024$ under the unit-normalised convention of Eq. (10). The Phase 05 value $L_{\text{crit}} = 2.79$ characterises the full production attractor under chaotic forcing, where $\lambda_{\max} = 0.3578$. The two measurements are complementary: Phase 03 measures predictability of the coupled latent dynamics under mild perturbation, whereas Phase 05 measures the intrinsic chaotic horizon of the production attractor.

The symplectic determinant $\det(W) = 1$ is verified analytically (Section 4.3) and confirmed numerically; the dissipation bound $\langle F, p \rangle \leq 0$ holds exactly under the

branchless Rayleigh damping update (11). The fourth-order vorticity estimator used by the vorticity preservation score (VPS) is

$$\begin{aligned} D_x u_{i,j}^y &= \frac{-u_{i+2,j}^y + 8u_{i+1,j}^y - 8u_{i-1,j}^y + u_{i-2,j}^y}{12h}, \\ D_y u_{i,j}^x &= \frac{-u_{i,j+2}^x + 8u_{i,j+1}^x - 8u_{i,j-1}^x + u_{i,j-2}^x}{12h}, \\ \omega_{i,j} &= D_x u_{i,j}^y - D_y u_{i,j}^x + \mathcal{O}(h^4). \end{aligned} \quad (13)$$

with $VPS = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} (\omega_{i,j}^{\text{model}} - \omega_{i,j}^{\text{ref}})^2$. The reported value 1.25×10^{-9} confirms that the latent velocity field remains within physically acceptable vorticity bounds for this rollout. Memory constraints are satisfied: both the 5 GB total RSS and 1 GB streaming-buffer ceilings are respected (Table 3).

5.4 Phase 04: Production Execution and Stability Profiling

We scale to 524288 simultaneous trajectories across 512-dimensional phase space (total element count: 268 435 456), orchestrated via a Python subprocess wrapper that queries `RUSAGE_CHILDREN` to correctly attribute the C++ worker memory.

Table 4: Phase 04 production metrics.

Metric	Value
Throughput (steps/sec)	1.2×10^6
Total fractional drift ($ \Delta E /E_0$)	1.26×10^{-11}
Linear response error	$< 1\%$
Peak worker RSS	4096.50 MB
Peak orchestrator RSS	10.51 MB

A real-time monitoring loop queries `getrusage` every 10^4 steps and enforces a soft-ceiling abort at 4.5 GB, leaving margin for OS fluctuations. Chaos injection perturbs the momentum field with $\delta p \sim \mathcal{U}(-\varepsilon, \varepsilon)$ using a spatially correlated kernel $\exp(-|x - x_c|^2/\sigma^2)$ to maintain VPS conditions; the Lyapunov divergence $\delta(t) = |x_1(t) - x_2(t)|$ is tracked against twin trajectories. Linear response checks every 10^5 steps confirm that small impulse deviations match the Jacobian prediction $J(t)\delta p$ within 1%.

5.5 Phase 05: Topological Reconstruction and Lyapunov Analysis

Phase 05 applies a full statistical pipeline to the unpadded, 64-byte-aligned binary telemetry from Phase 04.

Embedding. The embedding delay $\tau = 1$ is selected from the first local minimum of Average Mutual Information (AMI), reducing delay-coordinate redundancy.

False Nearest Neighbour analysis selects embedding dimension $m = 3$. The delay-embedded attractor $X_i = [x_i, x_{i+\tau}, x_{i+2\tau}]$ is constructed for topological analysis.

Ablation: Symplectic vs. Non-Symplectic Integration

Table 5 compares ECE at four checkpoints over 10^4 integration steps on a unit harmonic oscillator ($\omega = 1$, $\Delta t = 0.01$, initial conditions $q_0 = 1$, $p_0 = 0$).

Table 5: ECE checkpoints over 10^4 steps ($\omega = 1$, $\Delta t = 0.01$). Euler ECE grows as $e^{N\omega^2\Delta t^2} - 1$; Störmer–Verlet ECE remains bounded and oscillatory.

Step	Euler	Verlet	Ratio
100	1.005×10^{-2}	1.770×10^{-5}	$568 \times$
1 000	1.052×10^{-1}	7.400×10^{-6}	$14,216 \times$
5 000	6.487×10^{-1}	1.718×10^{-6}	$377,590 \times$
10 000	1.718×10^0	6.401×10^{-6}	$268,400 \times$

Euler (non-symplectic). Forward Euler produces a non-symplectic map whose Jacobian satisfies $\det(M) = 1 + \omega^2\Delta t^2 > 1$, expanding phase-space volume at every step. Energy grows monotonically as $(1 + \omega^2\Delta t^2)^N - 1 \approx e^{N\omega^2\Delta t^2} - 1$, reaching ECE = 1.718 at step 10^4 —a divergence that renders long-horizon rollouts physically meaningless.

Störmer–Verlet (PG-S3). The symplectic map conserves a shadow Hamiltonian $\tilde{H} = \mathcal{H} + O(\Delta t^2)$ [15], bounding the ECE by $\omega^2\Delta t^2/4 = 2.50 \times 10^{-5}$. Crucially, the observed ECE is not a fixed offset but *oscillatory*, varying between 1.7×10^{-6} and 1.8×10^{-5} across 10^4 steps with no secular growth. This behaviour is the direct signature of bounded motion on a shadow energy surface rather than outward spiralling, confirming that PG-S3 preserves the topological structure of Hamiltonian phase space. At step 10^4 the improvement over Euler is $268,400 \times$, a ratio that grows without bound as $N \rightarrow \infty$ because Euler diverges while PG-S3 remains bounded. This is consistent with the production energy drift of $1.26 \times 10^{-11} E_0$ observed in Phase 04, where the Gershgorin CFL condition adaptively contracts Δt to $\approx 5 \times 10^{-4}$, further suppressing the shadow Hamiltonian offset.

Lyapunov Exponent. The Rosenstein algorithm [13] tracks the average divergence of nearest-neighbour pairs over a 100-step window. Log-linear regression on the divergence curve yields:

$$\lambda_{\max} = 0.3578, \quad R^2 = 0.9792, \quad (14)$$

indicating high-confidence convergence within the chaotic growth regime. In Phase 05, we adopt the simplified form $L_{\text{crit}} = 1/\lambda_{\max}$, corresponding to the unit-normalised threshold $\ln(\varepsilon/\|\delta h(0)\|) = 1$, i.e. an initial separation of $\|\delta h(0)\| = \varepsilon/e$. This convention fixes the predictability scale to the natural unit of the embedding time step. The predictability horizon $L_{\text{crit}} = 1/\lambda_{\max} \approx 2.79$ (in units of the embedding time step).

Poincaré Section. Adaptive sectioning at $z = \text{mean}(X[:, 2])$ captures manifold intersections without bias from fixed-plane methods.

Correlation Dimension. The correlation integral $C(r) \propto r^{D_2}$ is estimated using a KD-tree nearest-neighbour approach over a dynamic scaling interval $r \in [r_{\min}, r_{\max}]$, yielding $D_2 = 2.12$, consistent with a dissipative strange attractor of low topological complexity.

Table 6: Phase 05 topological invariants and performance.

Metric	Value	Status
$\lambda_{\max}$ (Lyapunov)	0.3578	Verified
Convergence R^2	0.9792	High confidence
L_{crit} (horizon)	2.79	Consistent
Correlation dim. D_2	2.12	Consistent
Peak analysis RSS	4096.5 MB	PASS
Ingestion throughput	> 500 MB/s	PASS

6 Discussion

Symplecticity and energy drift. The reported energy-drift metric is the total fractional Hamiltonian drift $|\Delta \mathcal{H}|/E_0$ accumulated over the full production run of $N = 1.2 \times 10^9$ integration steps. This is a decisive improvement over non-symplectic baselines, which can exhibit secular energy growth even when their local truncation error is small [15]. The oscillatory—rather than monotonically growing—nature of the Störmer–Verlet ECE (Table 5) provides direct empirical evidence that the numerical trajectory remains on a nearby shadow energy surface, the geometric property underlying the long-horizon stability of PG-S3.

Predictability horizon. The empirically measured $L_{\text{crit}} = 2.79$ characterises the horizon of the particular attractor probed in Phase 05. For smoother Hamiltonian dynamics with smaller $\lambda_{\max}$, L_{crit} scales inversely, meaning that physical environments with weaker chaos (e.g., near-integrable rigid bodies) will exhibit substantially longer predictability.

Limitations. The Phase 02 oscillator is a controlled unit test of the integrator, not a substitute for full physical validation. Phase 04 therefore evaluates the production core on trajectory ensembles and attributes worker memory via `RUSAGE_CHILDREN`. As an architectural roadmap, the current coupling layer uses fixed scalar shear parameters α, β, γ to keep the phase-space transform analytically symplectic. A natural next extension is to make these parameters learnable while constraining the update to remain in $Sp(d, \mathbb{R})$, enabling data-adaptive symplectic manifolds without giving up the geometric guarantee. Additionally, the 4-bit SAQ bound assumes locally uniform quantisation noise; learned or information-aware 4-bit formats such as NF4-style quantisation [16] could improve spectral tail fidelity further.

7 Conclusion

We presented PG-S3, a world-model architecture that marries the $O(L)$ complexity of selective state space models with the geometric integrity of Hamiltonian mechanics. By replacing ZOH discretization with a Störmer–Verlet symplectic integrator, enforcing a Gershgorin CFL stability bound, and deploying SAQ INT4 quantization on AVX2 hardware, PG-S3 achieves: (i) Hamiltonian energy drift of $1.26 \times 10^{-11} E_0$ at 1.2×10^6 steps/second on consumer CPU hardware; (ii) a verified memory footprint of 4 096.5 MB for 524 288-trajectory production runs; and (iii) topological reconstruction of the learned latent attractor with $\lambda_{\max} = 0.3578$ and $L_{\text{crit}} = 2.79$. These results confirm that physical conservation laws and hardware-level efficiency are not in tension—they are mutually reinforcing constraints that, together, enable robust, long-horizon physical world modeling on accessible hardware.

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